Supplementary Materials

Cross-Cultural Validation and Ideological Fairness of a Historical Perspective Taking Instrument: Evidence from Czech Secondary Students

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Table S1: GRM Item Parameters

Graded Response Model (GRM) item discrimination (a_1) and category-intercept (d_1, d_2, d_3) parameters for the nine HPT items. Parameters were estimated using the `mirt` R package with the constrained multi-group model (`mod_base`; High- vs. Low-ideology groups defined by tertile split on the composite FR-LF + KSA score). Nine joint omnibus item tests released each item's slope and threshold constraints. Bonferroni-adjusted p-values (raw $p \times 9$, capped at one) were compared with a familywise α of .05 (equivalent per-test $\alpha = .0056$); no item was flagged. Because no items were flagged, the constrained model—in which item parameters are held equal across groups—is the final model. The parameters below therefore apply to both ideology groups.

Item	Subscale	a_1	d_1	d_2	d_3
CONT1	Contextualization	1.853	2.801	0.733	-1.958
CONT2	Contextualization	1.414	2.098	0.728	-1.311
CONT3	Contextualization	1.359	2.755	0.755	-1.319
POP1	Present-oriented perspective	-1.017	0.198	-1.369	-2.671
POP2	Present-oriented perspective	-0.351	0.896	-0.860	-2.210
POP3	Present-oriented perspective	-0.643	0.702	-1.063	-2.728
ROA1	Role of historical agent	1.181	2.620	1.192	-0.796
ROA2	Role of historical agent	0.708	2.148	0.492	-1.770
ROA3	Role of historical agent	1.416	2.690	1.297	-0.946

Note. a_1 = item discrimination; d_1, d_2, d_3 = category intercepts in `mirt`'s default parameterization. Raw POP responses entered the GRM and were not reversed for this analysis; their negative discrimination estimates therefore reflect the opposite direction of the present-oriented items relative to CONT and ROA.

POP was reversed only when computing composite scores. The constrained model was retained because the joint omnibus likelihood-ratio DIF tests showed no significant difference between free and constrained parameters for any item (all adjusted $p > .05$).

Table S2: HPT Standardized Factor Loadings

Item	Subscale	Loading	SE	95% CI
POP1	Present-oriented perspective	.74	.099	[.54, .93]
POP2	Present-oriented perspective	.30	.076	[.15, .44]
POP3	Present-oriented perspective	.50	.068	[.37, .63]
ROA1	Role of historical agent	.61	.071	[.47, .75]
ROA2	Role of historical agent	.30	.074	[.15, .44]
ROA3	Role of historical agent	.67	.074	[.53, .82]
CONT1	Contextualization	.72	.055	[.61, .82]
CONT2	Contextualization	.61	.059	[.49, .72]
CONT3	Contextualization	.64	.054	[.53, .74]

Note. POP items are reverse-scored. Estimates are from the three-factor WLSMV model. All loadings were significant at $p < .001$.

Table S3: Reliability Estimates

		Complete			Polychoric	
Scale	Items	n	Raw α	95% CI	α	ω_t
HPT: reversed present-oriented perspective	3	282	.46	[.35, .57]	.52	.54
HPT: role of the historical agent	3	281	.50	[.40, .60]	.53	.55
HPT: contextualization	3	287	.64	[.57, .71]	.69	.70

Scale	Items	Complete n	Raw α	95% CI	Polychoric α	ω_t
HPT composite (six items)	6	282	.60	[.53, .68]	.65	.66
HPT total	9	276	.66	[.60, .72]	.69	.70
Historical knowledge	6	286	.55	[.46, .63]	-	-
FR-LF: dictatorship acceptance	3	279	.51	[.42, .61]	.56	.61
FR-LF: Nazi-crime relativization	3	280	.59	[.51, .68]	.63	.64
FR-LF total	6	275	.67	[.61, .73]	.70	.71
KSA-3: aggression	3	279	.61	[.53, .69]	.64	.66
KSA-3: submission	3	277	.39	[.26, .51]	.42	.44
KSA-3: conventionalism	3	276	.56	[.47, .65]	.60	.61
KSA-3 total	9	267	.72	[.67, .77]	.75	.75

Scale	Items	Complete n	Raw α	95% CI	Polychoric α	ω_t
Social desirability	5	279	.30	[.17, .43]	.34	.55
Combined ideology composite	2 scales	283	.80	[.74, .84]	-	-

Note. The historical-knowledge coefficient is KR-20, equivalent to raw alpha for dichotomous items. Alpha confidence intervals use the standard-error approximation reported by `psych::alpha`. The combined ideology coefficient is Mosier reliability for the mean of standardized FR-LF and KSA-3 scores; its interval is a percentile bootstrap interval with 2,000 resamples. The other omega estimates are total omega from a one-factor model of the polychoric matrix. For the multidimensional HPT, FR-LF, and KSA composites, these omega values are descriptive one-factor estimates rather than reliability derived from the correlated-factor measurement model.

Table S3b: Pairwise Sample Sizes for Table 5

	HPT composite	Context	Pres. (rev.)	Knowledge	FR-LF	KSA-3	SDR-5
HPT composite	-	287	287	287	283	282	282
Context	287	-	287	287	283	282	282
Pres. (rev.)	287	287	-	287	283	282	282
Knowledge	287	287	287	-	284	283	283
FR-LF	283	283	283	284	-	283	283
KSA-3	282	282	282	283	283	-	283
SDR-5	282	282	282	283	283	283	-

Table S4: Instrument Source Populations and Language

Instrument	Source population	Source language	Relevance to this sample
HPT	170 German Grade 10 students; Dutch extensions included ages 10-17	German; published validation in English; later Dutch translation	Direct secondary-student evidence exists, but only in German and Dutch contexts
FR-LF mini	German population surveys sampled respondents aged 14 years and older; samples were predominantly adult	German	Includes adolescents but provides no distinct adolescent or Czech validation
KSA-3	German-speaking general population aged 18 years and older; development samples $n = 228$ and 223, mean ages 51.4 and 49.0	German	No adolescent or Czech validation in the source report
SDR-5	US medical outpatients ($n = 614$, mean age 37; $n = 3{,}053$, mean age 47) and 75 older adults	English	No adolescent or Czech validation in the source report
Historical knowledge	Six study-specific items for the Weimar scenario	Czech	Scenario-proximal criterion rather than a general knowledge scale

The non-HPT scales were administered in Czech. Beyond expert review of the battery, separate cognitive-interview or psychometric validation studies of these Czech adolescent versions were not available. Their use should therefore be interpreted through the sample-specific reliability estimates in Table S3 rather than assumed population portability.

Table S4b: Analysis-Specific Sample Sizes

Analysis or score	Included n
Recruited sample	293
Three-factor CFA	276
HPT score descriptives	287
Historical-knowledge descriptives	293
FR-LF descriptives	284
KSA-3 descriptives	283
SDR-5 descriptives	283
Reliability	267-287, by complete item set (Table S3)
Pairwise correlations	282-287 (Table S3b)
Composite-ideology DIF groups	96 low, 95 high; 92 middle excluded; 10 missing ideology score
Composite-ideology MG-CFA	92 low, 90 high after listwise exclusion of incomplete HPT item responses
NS-only MG-CFA groups	126 low, 123 high; 37 middle excluded; 7 missing NS scores

The differences arise from item nonresponse, the scale-specific minimum-answer rules described in the manuscript, listwise-complete item sets for reliability, pairwise-complete correlations, and the extreme-group exclusions used for DIF and MG-CFA (registered for the MG-CFA sensitivity model).

Knowledge-item facilities were .57 (KN1), .44 (KN2), .73 (KN3), .44 (KN4), .41 (KN5), and .47 (KN6). The total score had $M = 3.04$, $SD = 1.62$, median = 3, IQR [2, 4], with 4.1% at zero and 8.9% at six.

Table S5: Preregistration Mapping and Deviations

The immutable OSF registration https://osf.io/zsngy was registered on 27 November 2025. The table records how the present report differs from that plan.

Registered element	Status in this report
H1: ideology positively predicts CONT with knowledge and SDR-5 covariates	Relevant associations are reported, but the exact registered one-tailed, covariate-adjusted test is not presented as a confirmatory test

Registered element	Status in this report
H2: ideology negatively predicts raw POP	Relevant associations are reported, but the exact registered one-tailed model is not presented as a confirmatory test. The observed KSA-3/reversed-POP direction was not predicted: H2 predicted the opposite direction on raw POP.
H3: ideology positively predicts total HPT with covariates	Related models are reported, but the six-item HPT composite was adopted post-registration as the primary outcome
H4: positive DIF on CONT items	The registered primary continuous-ideology MIMIC analysis was not implemented; a post-registration all-item GRM DIF analysis is reported instead. The registered tertile-group MG-CFA sensitivity model was implemented with the registered steps and criteria but extended from the CONT items to the full three-factor model
H5: stronger ideology-HPT associations at lower knowledge	Neither the FR-LF $\times$ knowledge nor KSA $\times$ knowledge moderation analysis, nor the registered simple slopes, was implemented
H6: exploratory positive correlations and FR-LF-CONT > KSA-CONT	Correlations are reported; class-clustered errors and the registered Williams/Steiger comparison were not implemented
Two-level class random-intercept models	Three-level and school-level specifications were added post-registration
Grade level, gender, history grade, knowledge, and SDR-5 covariates	The complete registered covariate set was not used in all reported models
One-tailed tests and Benjamini-Hochberg correction	Reported tests are two-tailed; no reported test implemented the registered Benjamini-Hochberg correction
Registered missing-data rule	The registered listwise-deletion/MICE decision rule was not implemented; the score-specific rules used in the manuscript are documented in Section 3.4

Registered element	Status in this report
HPT response scale 0-3; two ROA items; approximately ten knowledge items	Administered battery used 1-4 HPT responses, three ROA items, and six knowledge items
CFA-focused validation, NS-only invariance, TOST, and alternative exclusions	Added after registration and treated as exploratory or descriptive
Orthogonal bifactor CFA	Added post-registration. Its scaled global fit indices favor the more complex model, but the non-nested, parameterization-dependent comparison and three indicators per specific factor limit strong structural claims.

Table S6: NS-Only Multi-Group CFA Fit

Model	CFI	RMSEA	SRMR	ΔCFI	ΔRMSEA	ΔSRMR
Configural	.959	.051	.071	-	-	-
Metric	.961	.047	.076	+.002	-.004	+.005
Scalar	.976	.034	.073	+.015	-.013	-.003

Note. Groups were formed by a tertile-based split on the National Socialist sympathy/relativization (NS) facet of the FR-LF, unequal because of tied scores (126 low, 123 high; 37 middle and 7 missing excluded). Three-factor model, WLSMV estimator, items treated as ordered categorical. Scaled $\chi^2(48) = 62.77$, $\chi^2(54) = 68.15$, and $\chi^2(63) = 71.86$ for the configural, metric, and scalar models; CFI and RMSEA also use lavaan's scaled WLSMV convention. Changes are relative to the preceding model; positive ΔCFI and negative ΔRMSEA indicate better fit. Source: `verify_statistics.R`, Step 3. This analysis was added after registration and is descriptive.

Table S7: HPT Composite by Composite-Ideology Tertile

Ideology tertile	n	M	95% CI
Low	96	2.91	[2.80, 3.03]
Middle	91	2.86	[2.76, 2.96]
High	95	2.75	[2.63, 2.87]

Note. HPT composite = mean of reversed present-oriented perspective and contextualization scores (1-4). Tertiles of the composite ideology

score (mean of standardized FR-LF and KSA-3 scores) use the cut points of the DIF analysis, so the Low and High groups are the DIF groups. Students with both scores available ($n = 282$). Intervals are unadjusted t intervals that ignore school and classroom clustering. Tertiles are arbitrary cut-points of a continuous score; the continuous relationship is shown in Figure 1. Source: `revision_reporting_analyses.R` (`revision_ideology_tertile_means.csv`).